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MANAGEMENT REVIEW

Management without theatre.



STRATEGY

PILOTS AND EXPERIMENTS BEFORE THE BIG CHANGE

An idea that waited six months, how few tested ideas improve anything, the Model for Improvement, why effects shrink at scale, when to trust a number, and a card for planning a pilot.

12% more revenue at Bing from a small change to ad headlines, an idea that had waited more than six months for someone to test it.

KOHAVI & THOMKE, 2017

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Big changes are usually decided on conviction: a plan, a budget, a launch date. There is another way: try the change small first, against a comparison group and with a decision rule agreed before the start, so that the result decides instead of the loudest opinion.

At Bing, a small idea waited more than six months and then raised revenue by 12%. At Microsoft, only one in three tested ideas improved what it was meant to improve. The Model for Improvement tests a change on a small scale: one day, one unit. In 126 government trials, effects were far smaller than in published studies. Four checks show when to trust a result, and a card helps plan the pilot.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure
Every figure has its source and year at the foot of its page.

Our reading
Where the editors interpret rather than the research, it says so.

Practice
The steps and the card are proposals to try, not research results.

COVER STORY

Six months in a drawer

In 2012 a Microsoft employee working on the Bing search engine proposed changing how ad headlines were displayed. It would take an engineer a few days, but it was one of hundreds of ideas, and the program managers rated it low priority.

The idea waited more than six months. Then an engineer, seeing how little the code would cost, launched a simple A/B test: users were assigned at random to the old headlines or the new ones. Within hours revenue rose so fast that it set off a “too good to be true” alert. Such alerts usually signal a bug. This time they did not.

6+
months the idea waited, rated low priority

12%
more revenue, without hurting key user-experience metrics

$100M+
a year in the US alone, by the authors' estimate

OUR READING

No meeting could have ranked this idea correctly. A cheap test answered in hours what months of setting priorities had not.

Source: Ron Kohavi & Stefan Thomke, Harvard Business Review 95(5), 2017

Kohavi and Thomke call it the best revenue-generating idea in Bing's history. The 12% was measured in the test; the $100 million is their annual figure for the US.

THE NUMBERS

Most ideas do not work

Companies that test almost every change learn the same lesson, write Ron Kohavi and Stefan Thomke: most new ideas fail in experiments, and even experts often misjudge which ones will pay off. At Microsoft as a whole:

IDEAS TESTED IN EXPERIMENTS AT MICROSOFT, BY RESULT



1 in 3
prove effective

1 in 3
have neutral results

1 in 3
have negative results

In well-optimised products such as Google and Bing, only about 10–20% of experiments give positive results; at Slack, about 30% of monetisation experiments (2019). Thomke puts the average at about 10%.

OUR READING

If two in three plausible ideas change nothing or make things worse, rolling out a big change untested amounts to a bet.

Sources: Ron Kohavi & Stefan Thomke, Harvard Business Review 95(5), 2017; Ron Kohavi, Diane Tang & Ya Xu, Cambridge University Press, 2020 (via Chapter 1, published by the authors); Stefan H. Thomke, Harvard Business Review Press, 2020 (via Michael Blanding, “Creating the Experimentation Organization”, HBS Working Knowledge, 2019)

Online products, where tests are cheap; for shop floors we found no comparable shares. Slack: a post quoted by Kohavi, Tang & Xu; Thomke's 10%: an interview (2019).

THE MODEL

Three questions, small cycles

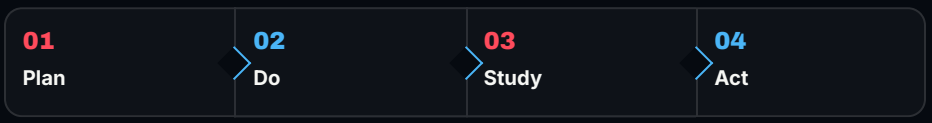
The Model for Improvement, from Associates in Process Improvement and set out in The Improvement Guide (Langley and colleagues, 2009), asks three questions before anything is changed for good.

01
Aim
What are we trying to accomplish?

02
Measure
How will we know that a change is an improvement?

03
Change
What change can we make that will result in improvement?

TESTING THE CHANGE: THE PLAN-DO-STUDY-ACT CYCLE



Plan the change, try it, study the results, act on what was learned. The test is small on purpose: the Institute for Healthcare Improvement's example is one patient, one day. After several cycles under varying conditions, such as morning and night, the change is made permanent and spread.

OUR READING

A pilot is not a soft launch. It is a question with a deadline: does this change, here, do better than what we do now?

Source: Langley, Moen, Nolan, Nolan, Norman & Provost, Jossey-Bass, 2009 (via Institute for Healthcare Improvement, “Model for Improvement”)

The questions and steps follow the Institute for Healthcare Improvement's page on the model; the book itself was not seen. Any interpretation beyond that is ours.

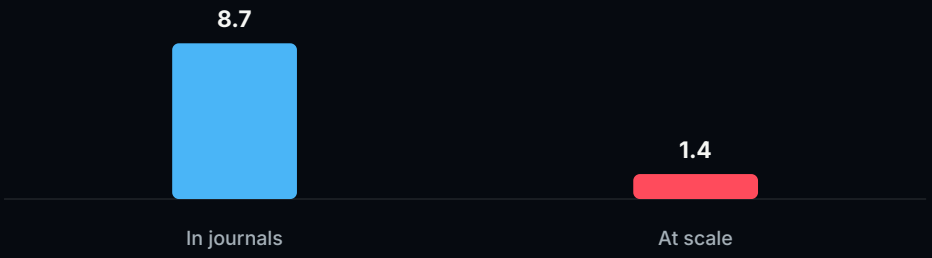
More in the essay DMAIC without a black belt

WHAT THE RESEARCH SAYS

Why results shrink at scale

Stefano DellaVigna and Elizabeth Linos gathered 126 randomised trials with 23 million people, all run by two large “nudge units” in the United States, and compared them with nudge trials published in academic journals.

AVERAGE EFFECT OF A NUDGE ON TAKE-UP, PERCENTAGE POINTS



The published trials were small: the median treatment group had 484 people, against 10,006 in the nudge units. Small trials can only detect large effects, and journals mostly print what is detected. Selective publication, made worse by low statistical power, explains about 70% of the gap; differences between the nudges explain most of the rest.

OUR READING

The success story you read about is the one that got published. Plan for the rollout to deliver less than the pilot that convinced you.

Source: Stefano DellaVigna & Elizabeth Linos, Econometrica 90(1), 2022 (via The authors' version, April 2021)

Take-up: the share who do what is asked, such as enrolling or getting vaccinated. Effects and the 70% from Econometrica (2022); group sizes from the 2020 working paper.

HOW IT IS MEASURED

Numbers you can trust

“Getting numbers is easy; getting numbers you can trust is hard,” write Kohavi and Thomke. Four checks before a pilot result decides:

1. **Compare with a control group**
   At Yahoo, an observational study put the effect of display ads on brand searches at 871–1,198%; a controlled experiment found 5.4%.

2. **Agree one main measure and guardrails**
   Before the start, not after; guardrails are what must not get worse.

3. **Run at least a full week**
   LinkedIn measures for at least a week: a single day over-represents heavy users.

4. **Check the split, doubt the surprise**
   Unplanned group sizes often void a result. Twyman's law: any figure that looks interesting or different is usually wrong.

HYPOTHETICAL EXAMPLE, A NEW PICKING ROUTE IN A WAREHOUSE

Comparison	zone A new route, zone B as before, the same two weeks
Main measure	lines picked per hour: A +6%, B +4%
Guardrail	picking errors: unchanged in both

Measured against zone B, the route is worth about 2 points, not 6. Invented figures, for illustration.

Sources: Ron Kohavi & Stefan Thomke, Harvard Business Review 95(5), 2017; Ron Kohavi, Diane Tang & Ya Xu, Cambridge University Press, 2020 (via Chapter 1, published by the authors); Ya Xu, Weitao Duan & Shaochen Huang, KDD '18 (ACM), 2018 (via The authors' preprint (arXiv), 2018)

Checks after Kohavi & Thomke (2017), Kohavi, Tang & Xu (2020) and, for the week, Xu et al. (2018). The example is the editors'.

TOOL OF THE ISSUE

The pilot plan card

Fill it in before the start, with those who will run the pilot. Then roll out in steps: a small group first, a measuring stage, then everyone.

01 Hypothesis if we change X, Y improves by about Z, because ...

02 What changes one change, so clear that anyone could repeat it

03 Where and how long one area, shift or team; at least a full normal week

04 What is measured one main measure, how it is counted, and the guardrails

05 Comparison group where the current way goes on, the same days

06 Decision rule set before the start: roll out, adjust or stop, and who decides

Sources: Langley, Moen, Nolan, Nolan, Norman & Provost, Jossey-Bass, 2009 (via Institute for Healthcare Improvement, “Model for Improvement”); Ya Xu, Weitao Duan & Shaochen Huang, KDD '18 (ACM), 2018 (via The authors' preprint (arXiv), 2018); Stefan H. Thomke, Harvard Business Review Press, 2020 (via Michael Blanding, “Creating the Experimentation Organization”, HBS Working Knowledge, 2019)

The card is the editors' proposal, drawing on the Model for Improvement, Xu et al. (2018) and Thomke's rule: a testable hypothesis, a control group, and abiding by the results.

Open the tool: Sigma & Control Chart stivencatalyst.com/tools/sigma-control-chart/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

The Surprising Power of Online Experiments
Ron Kohavi & Stefan Thomke, Harvard Business Review 95(5), 2017
https://hbr.org/2017/09/the-surprising-power-of-online-experiments

Trustworthy Online Controlled Experiments: A Practical Guide to A/B Testing
Ron Kohavi, Diane Tang & Ya Xu, Cambridge University Press, 2020 · via Chapter 1, published by the authors
https://experimentguide.com/

Experimentation Works: The Surprising Power of Business Experiments
Stefan H. Thomke, Harvard Business Review Press, 2020 · via Michael Blanding, "Creating the Experimentation Organization", HBS Working Knowledge, 2019
https://www.library.hbs.edu/working-knowledge/creating-the-experimentation-organization

The Improvement Guide: A Practical Approach to Enhancing Organizational Performance (2nd ed.)
Langley, Moen, Nolan, Nolan, Norman & Provost, Jossey-Bass, 2009 · via Institute for Healthcare Improvement, "Model for Improvement"
https://www.ihi.org/resources/how-improve-model-improvement

RCTs to Scale: Comprehensive Evidence from Two Nudge Units
Stefano DellaVigna & Elizabeth Linos, Econometrica 90(1), 2022 · via The authors' version, April 2021
https://doi.org/10.3982/ECTA18709

SQR: Balancing Speed, Quality and Risk in Online Experiments
Ya Xu, Weitao Duan & Shaochen Huang, KDD '18 (ACM), 2018 · via The authors' preprint (arXiv), 2018
https://doi.org/10.1145/3219819.3219875

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked "via". The editors' interpretation is marked "Our reading". Figures that could not be confirmed are not in the issue.



MANAGEMENT REVIEW

Management without theatre.

Every issue, one management question, checked against the best research.

ALL ISSUES

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